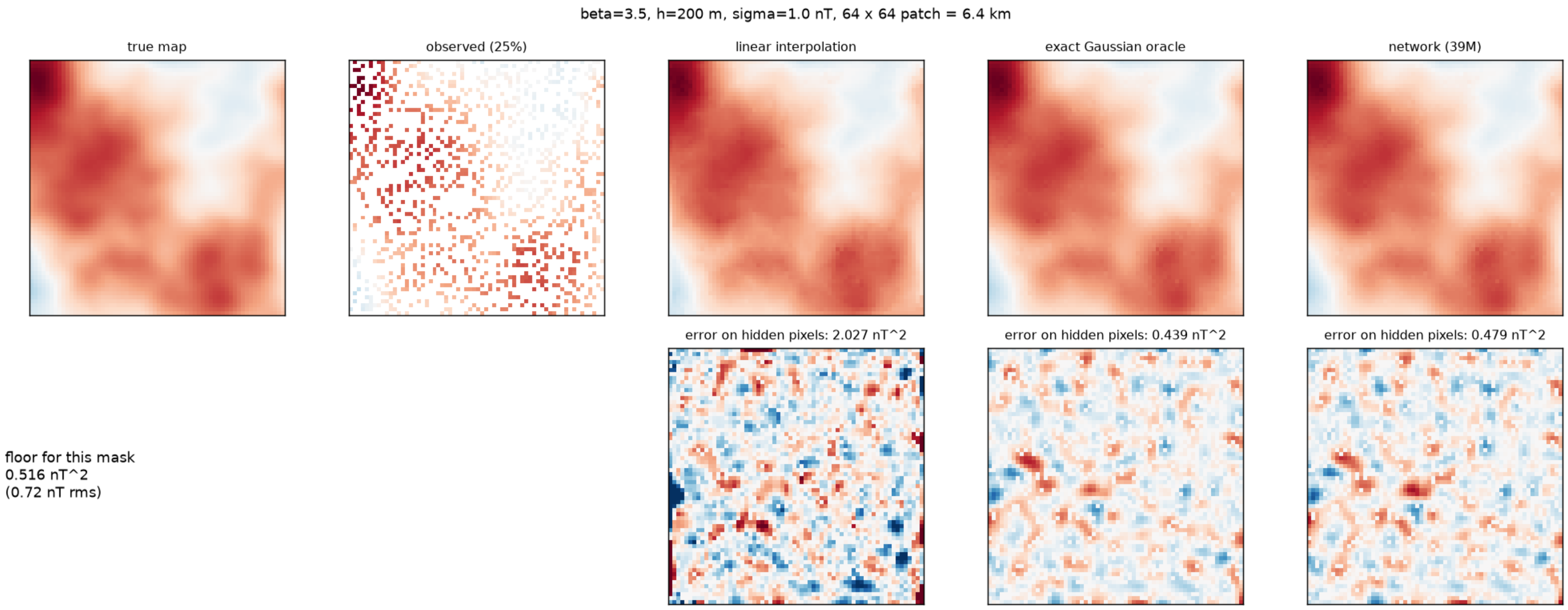
Reconstructing Magnetic Anomaly maps

Mehul Bandhu
Dirac Labs navigation research take-home, September 2026

Every number here is generated from committed run files in MehulBandhu/magnetic-navigation; the file behind each is in the notes.

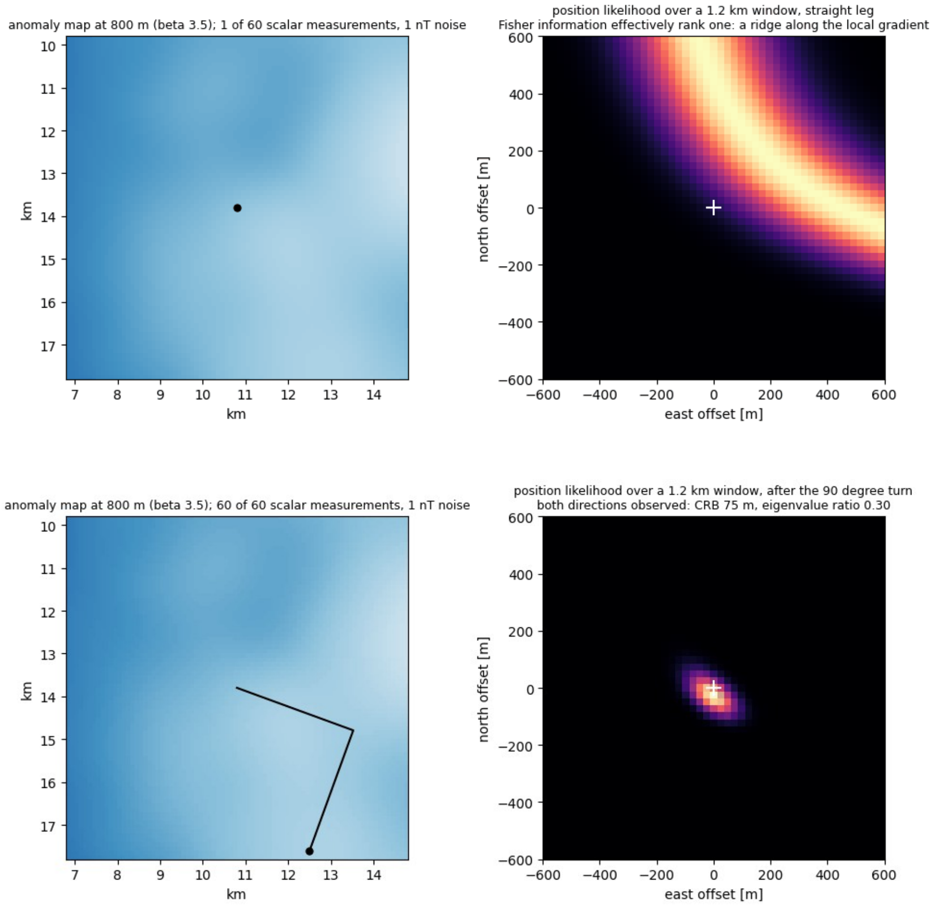
The maps are Gaussian random fields, so the best reconstruction has a closed form; every result is excess over that floor

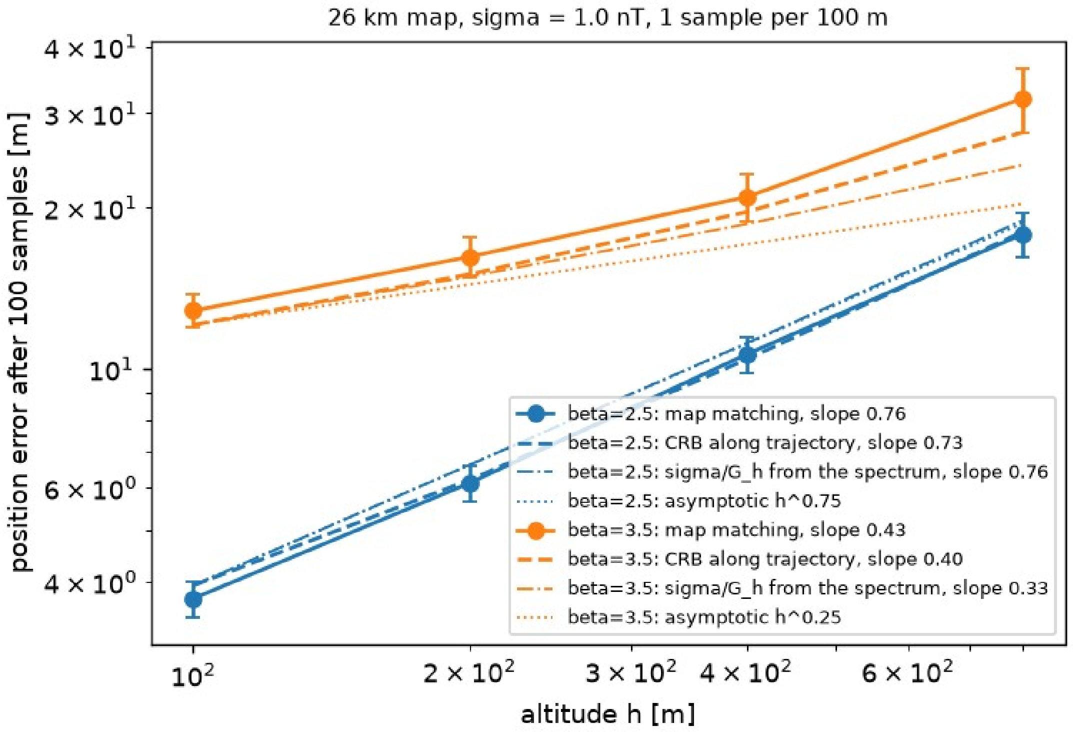


field P(k) ~ k^-beta e^-2kh → hide 75% of pixels, add 1 nT noise → network | exact estimator | interpolation → hidden-pixel error, minus the floor

One 6.4 km patch at 200 m: truth, the 25% observed, and three reconstructions with their hidden-pixel errors; the floor for this mask is 0.516 nT^2 (mask average 0.508).

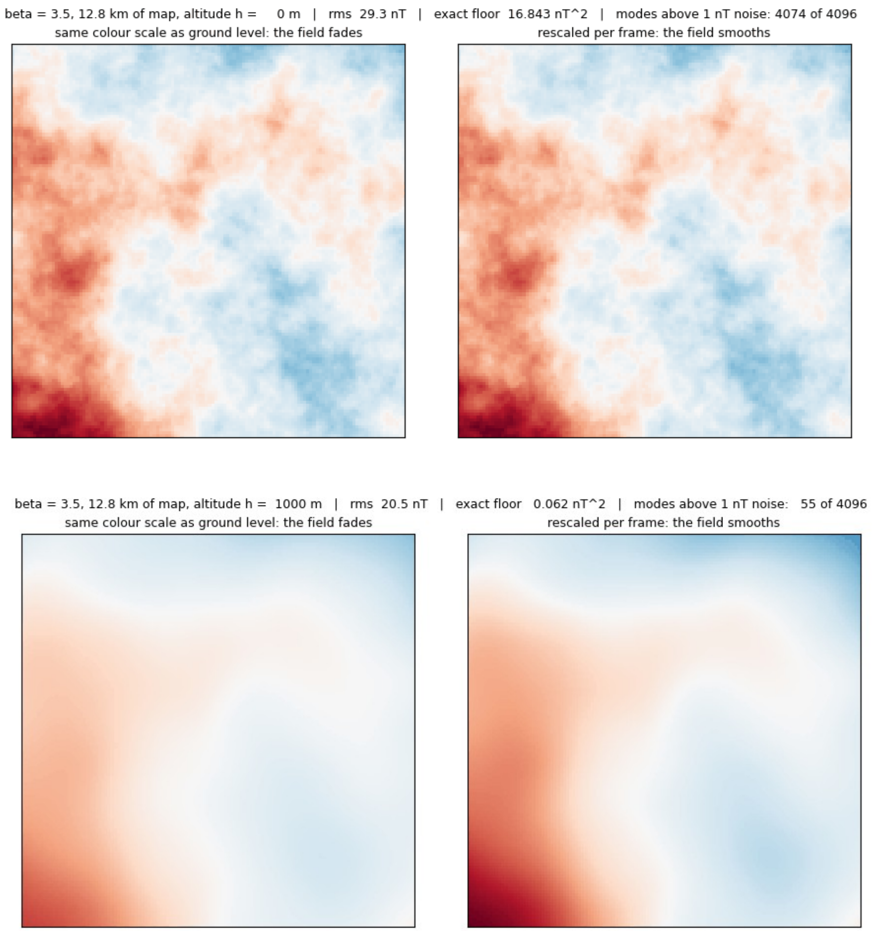
Problem 1: one scalar measurement fixes position along the gradient only; a turn makes it full rank; error grows as h^((4-beta)/2)

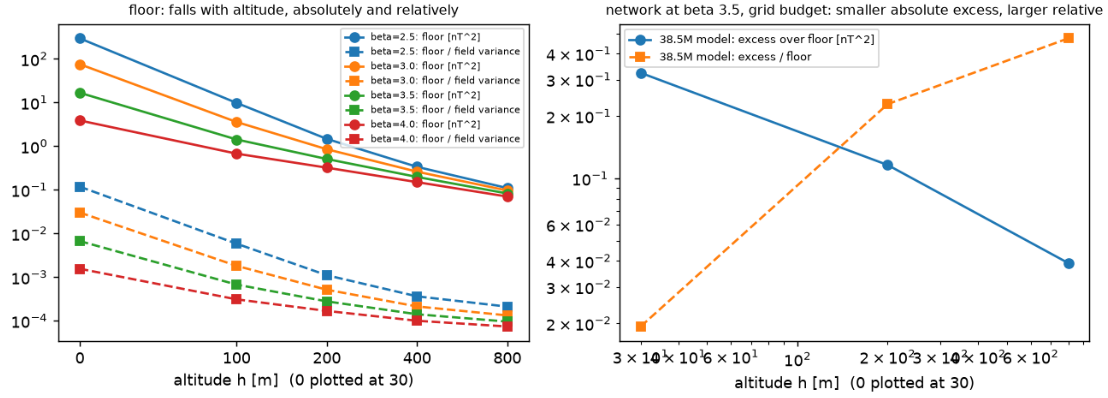




Left: the position likelihood after one measurement and after the turn (figures/navigation_likelihood.gif). Right: map-matching error against altitude on the Cramer-Rao bound: 0.76 against 0.75 at beta 2.5; at 3.5 the finite map bends the slope.

Altitude removes fine structure: the floor falls 200x from 0 to 800 m; easier in absolute error, harder relative to the floor

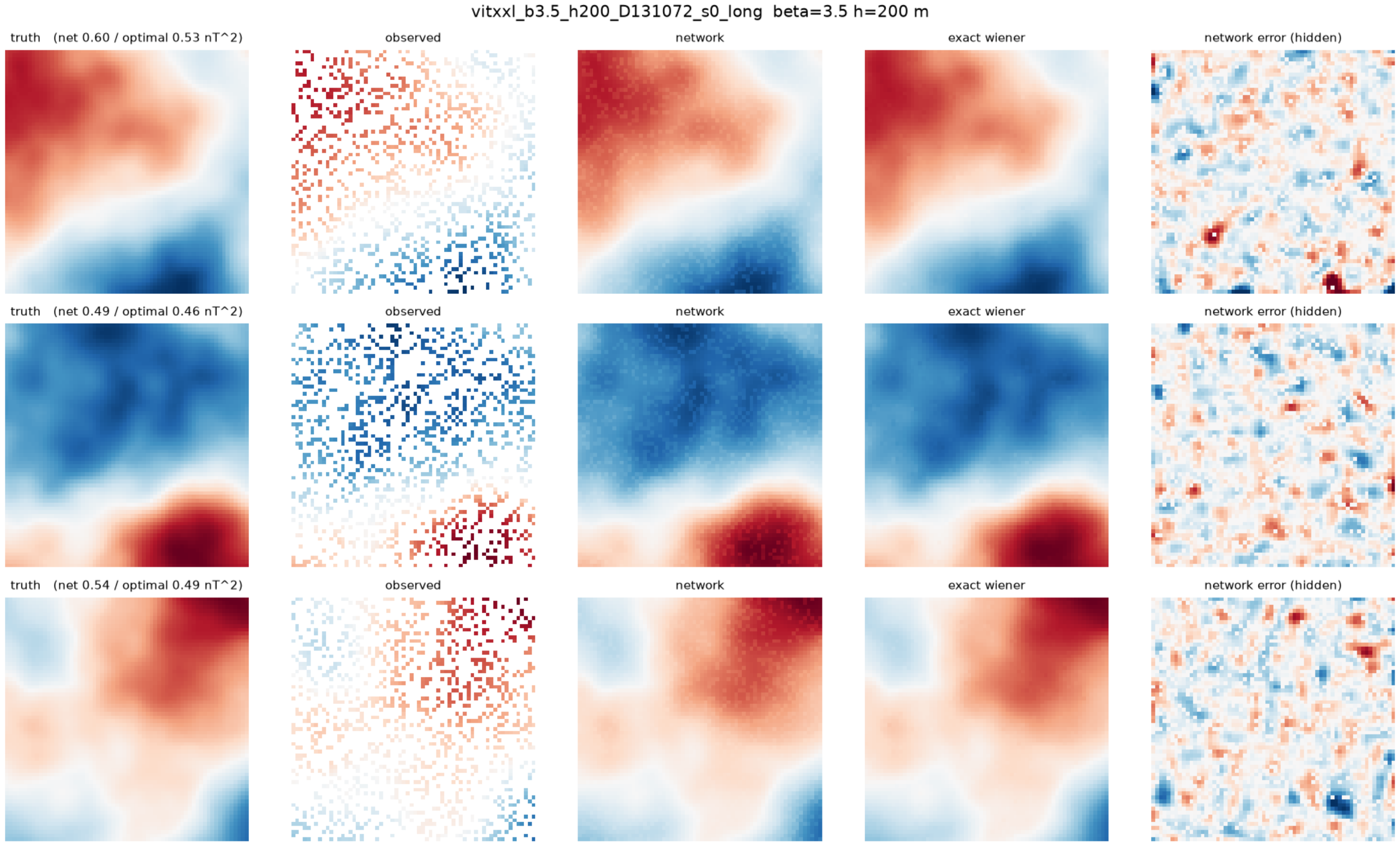




floor [nT^2], beta 3.5	**h=0**	**200**	**800**
	16.7	0.508	0.083
modes above 1 nT noise	4074	425	73

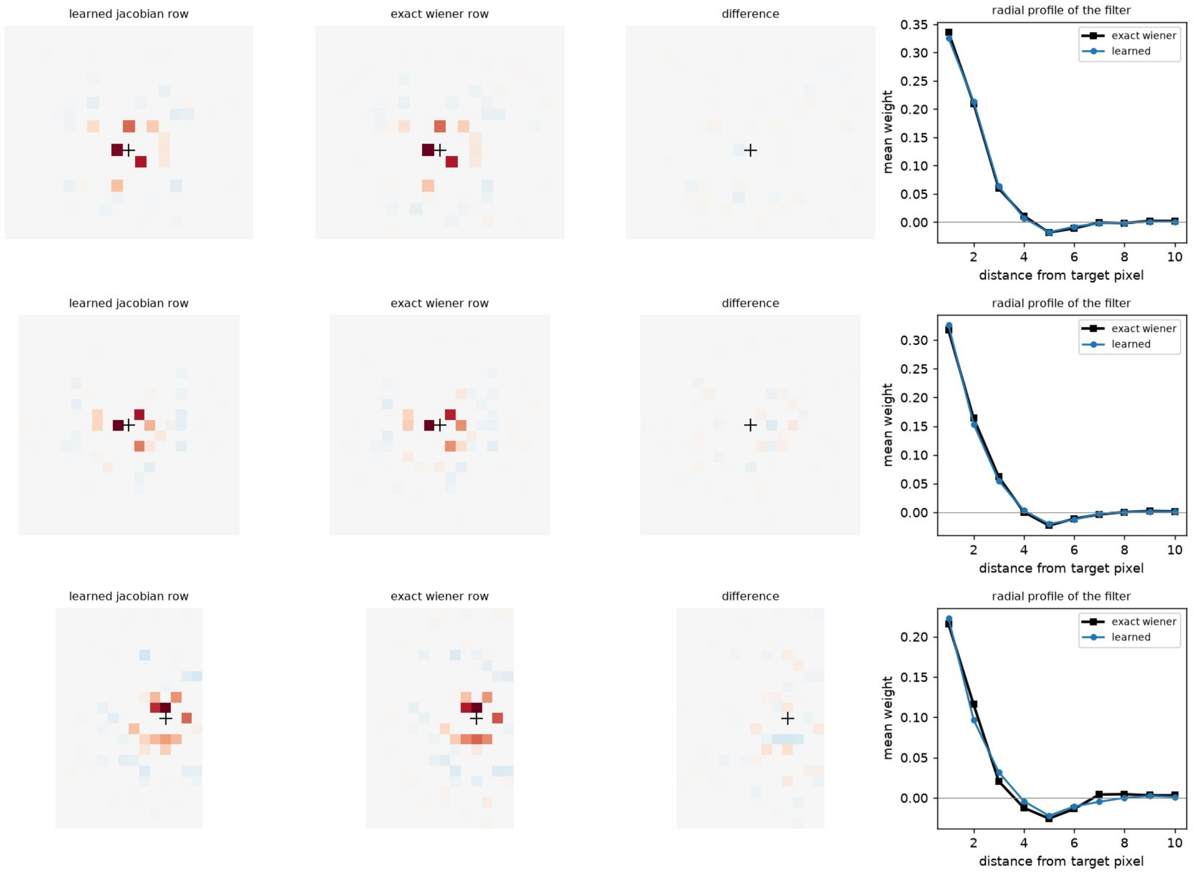
Left: the same field at 0 m and 1000 m (figures/upward_continuation.gif). Right: floor absolutely and relative to the field variance; the 38.5M model's excess absolutely and relative to the floor.

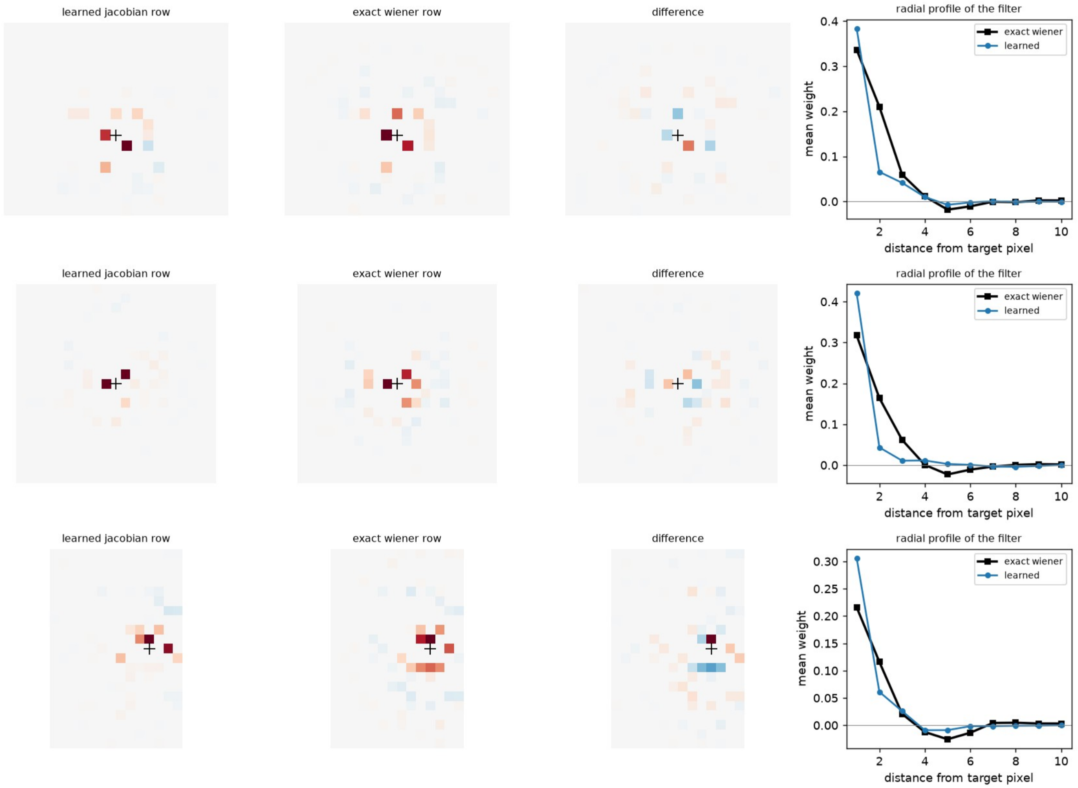
The 38.5M transformer reaches 0.553 after 24k steps against a floor of 0.508; the 8k-step reference run was at 1.10, the grid's 14k-step runs 0.62 to 0.65



Three patches at 200 m, 75% hidden: truth, observed, the exact estimator, the network after 24k steps, residuals. The 1.10 is a run whose schedule ended at 8k; the 24k run passing through step 8000 was at 0.876, a 21% schedule effect.

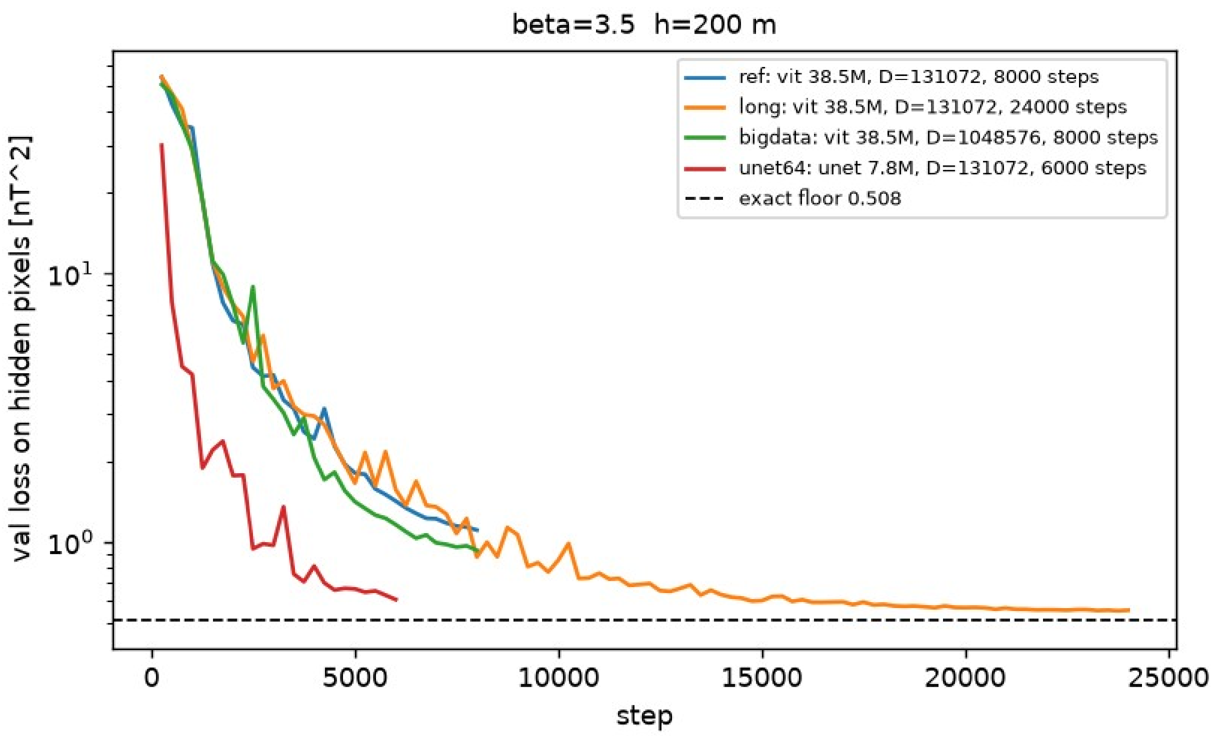
The trained network closely approximates the Wiener filter: linear to 0.15% at fixed mask after 24k steps, Jacobian rows within 16% of the exact rows





Rows of the input-output Jacobian at three hidden pixels against the exact Wiener rows for the same mask. Left: 24k steps, 16% error. Right: 8k steps, 59%.

Longer training reduces the gap from 0.60 to 0.045 in one run; on a fixed mask the model matches the exact estimator to 0.031 nT^2



target = exact conditional mean (no floor, fresh fields)

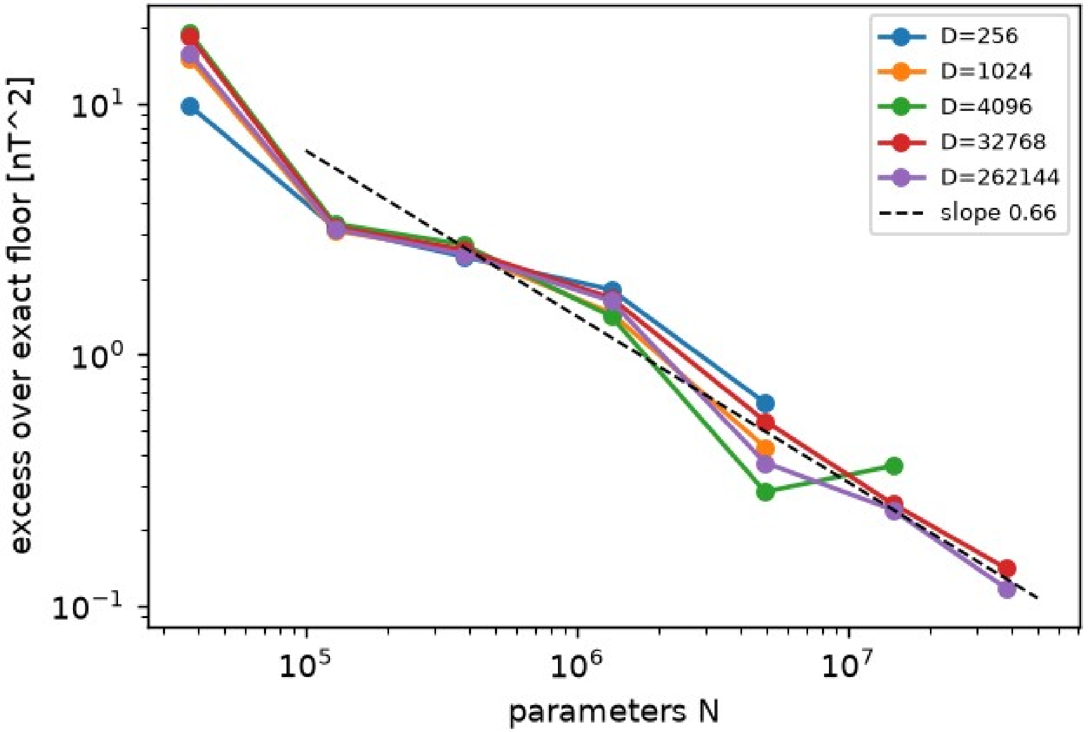
model	teacher masks	steps	distance to estimator
5M	1	8000	1.07
5M	1	24000	0.035
5M	8	24000	0.046
38.5M	1	24000	0.031

one run, 8k -> 24k steps: loss 0.876 -> 0.553

generalisation gap at 24k: 0.006

Left: validation loss against step for the 8k, 24k and 8x-data transformers and a 7.8M U-Net. Right: the teacher runs, whose loss has no floor, so the value after long training bounds the approximation error for those masks. The remainder on resampled masks is not decomposed further by this experiment.

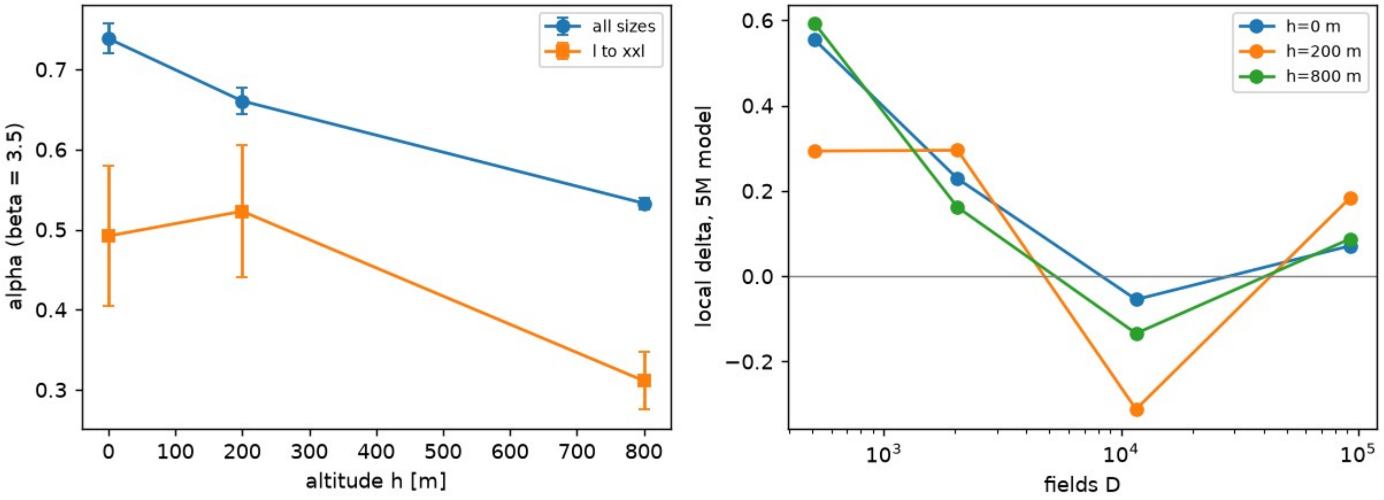
The parameter dependence is a step, not an exponent: 4 to 12x between 1.3M and 5M parameters, from width and training duration



one factor at a time, from the 1.3M model (excess 1.60)	
2000 more steps	1.03
width 256 at depth 4	1.09
depth 6 at width 160	1.84
head dimension 64	1.62
5M model (256 wide, 6 layers)	0.73 (seeds 0.31 to 0.73)

Left: error above the floor against model size, one line per dataset size (beta 3.5, 200 m); the drop between 1.3M and 5M appears at every condition. Right: five runs that change one thing at a time across that drop, at 32768 fields.

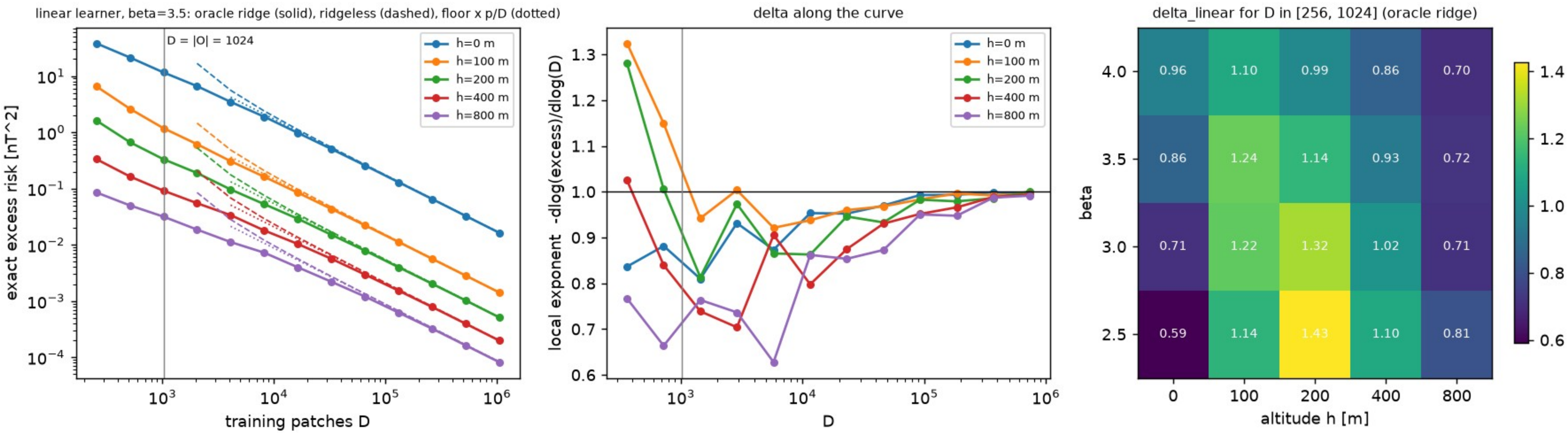
More than half of the apparent parameter scaling is the training budget growing with size: 0.66 across the grid, 0.25 at equal steps



condition	grid	equal steps	l..xxl
beta 2.5, 200 m	0.80	0.30	-0.01
beta 3.5, 0 m	0.74	0.37	0.01
beta 3.5, 200 m	0.66	0.25	-0.03
beta 3.5, 800 m	0.53	0.14	-0.28

Left: the grid slope against altitude, and the prediction that failed (steeper at altitude expected, flatter measured). Right: the same slope read from the logged curves at the same step for every size; above 5M parameters at equal steps there is no positive size dependence.

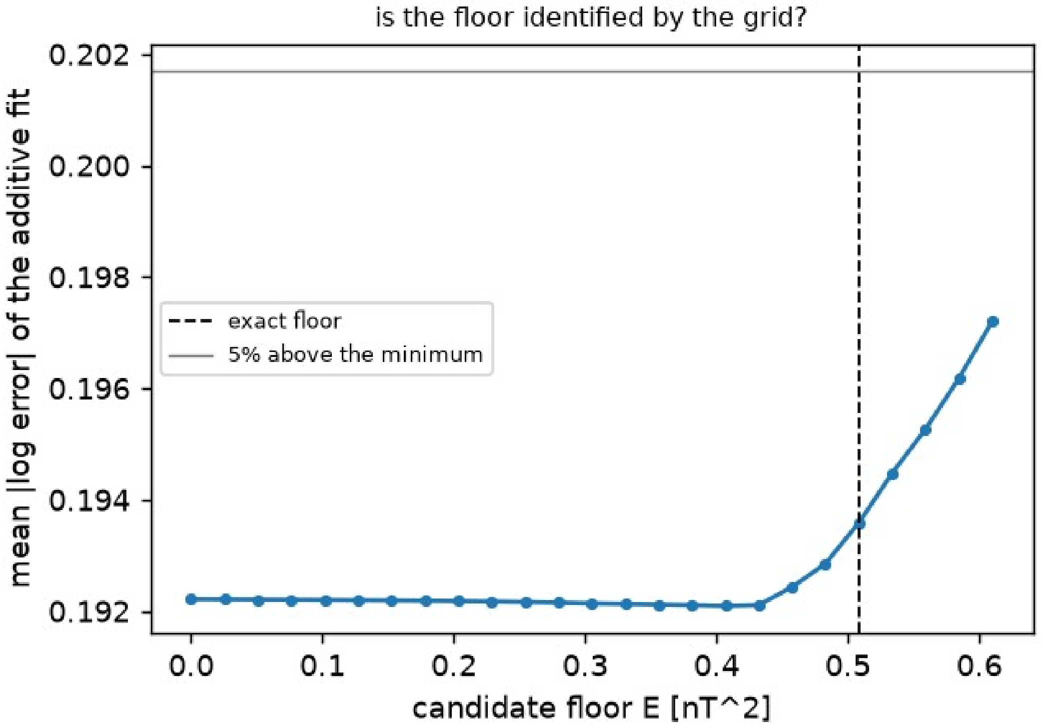
Beyond about a thousand maps, more data does not help



fixed mask, beta 3.5, 200 m: excess at 256 / 1024 / 4096 / 32768 fields
5M transformer 7.32 2.14 1.13 0.96 slope 0.89
linear regression 1.62 0.33 0.10 0.015 slope 1.14

The linear learner's excess against training patches at five altitudes, with the exact excess risk: slope 0.59 to 0.96 at ground level, above 1 near the effective rank, then p/(D-p-1). Below: the transformer on the identical fixed-mask task.

A fit with a free floor underestimates it by 5% to 65% and is nearly as good at any smaller value; the grid does not determine the floor, so the analytic one is used

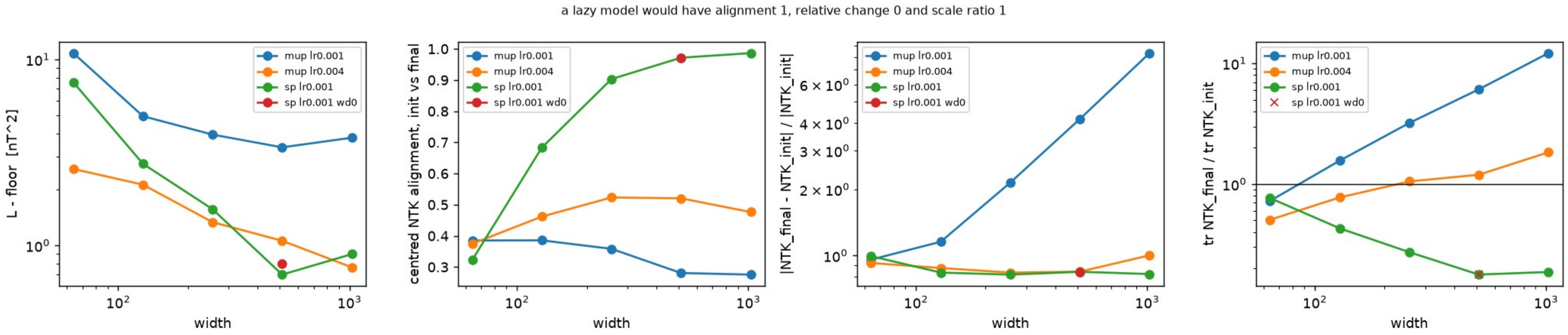


loss = E + A N^-alpha + B D^-delta, fitted and scored
in log total loss at every candidate E

condition	exact	free	ratio	E within 5% of best
beta 2.5, 200 m	1.469	0.51	0.35	0 to 1.03
beta 3.5, 0 m	16.73	14.2	0.85	1.7 to 15.1
beta 3.5, 200 m	0.508	0.41	0.80	0 to 0.61
beta 3.5, 800 m	0.082	0.078	0.95	0 to 0.10

Left: fit error against candidate floor at beta 3.5, 200 m; the exact floor is the dashed line. Right: the free-floor estimate at all four conditions. The earlier version scored log(loss - E), whose scale changes with E, and preferred zero; it was withdrawn.

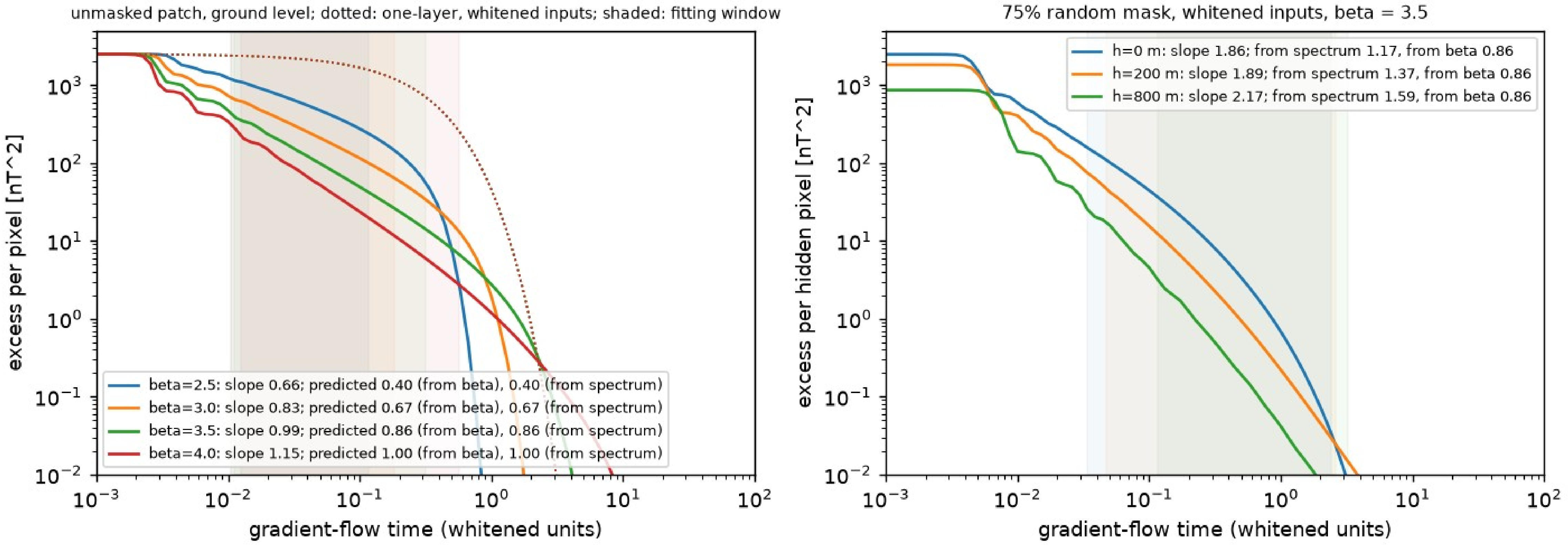
With standard scaling, wider models stop changing the direction of their kernel but shrink its size five-fold; with the muP-style scaling at lr 4e-3 the kernel keeps changing and the loss keeps falling with width



SP, width 64 -> 1024: centred alignment 0.32 -> 0.99 trace ratio 0.77 -> 0.19
width 512 without weight decay: alignment 0.971 vs 0.971, trace ratio 0.178 vs 0.178
muP-inspired, lr 4e-3: alignment 0.37 to 0.52, trace ratio rising to 1.8

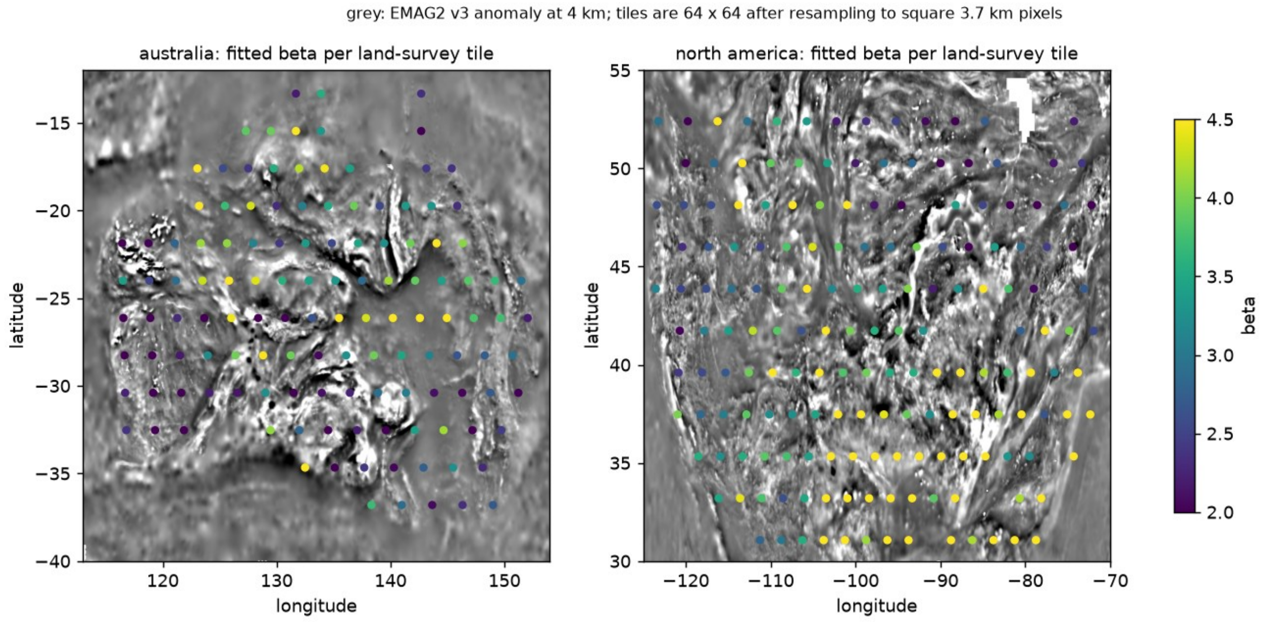
Loss, centred NTK alignment, relative change and trace ratio against width (depth 4, 4000 steps). A lazy model would sit at alignment 1, change 0, ratio 1. Neither parametrisation wins at matched width under this budget.

For a two-layer linear network the learning-curve exponent computed exactly agrees with the derived one up to the logarithm the derivation drops; with a mask the exponent changes, so the result does not carry over



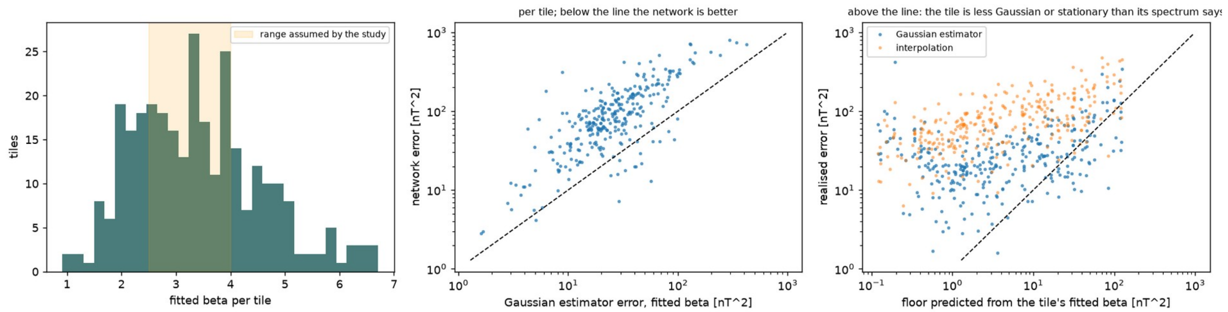
Closed-form gradient flow; every mode is a logistic. Left: ground level, four values of beta, slopes 0.66 to 1.15 against 0.40 to 1.00 predicted. Right: with a 75% mask the whitened spectrum steepens and the exponent with it.

On real anomaly maps the network does not transfer: worse than interpolation, 4x worse than the estimator with its own training prior



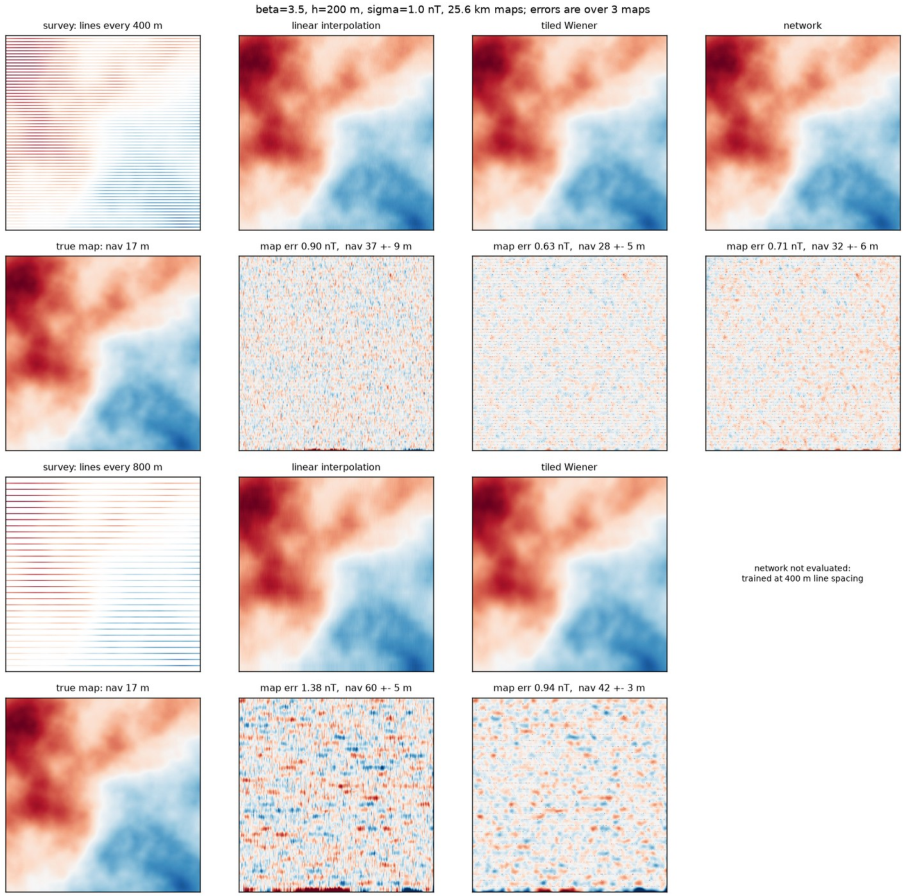
EMAG2 v3, 300 land tiles, h/dx = 1.08

	Australia	N. America	Gaussian tiles
fitted beta	2.9	3.5	
estimator / own floor	2.4	9.8	1.1
network / estimator	3.6	4.1	1.3 to 1.9
network nonlinearity	11%	12%	1 to 4%



Top left: fitted spectral slope per land-survey tile; the slope varies by geological province. Bottom left: slopes, network against estimator per tile, realised against predicted error of the estimator.

Map error becomes position error: 28 m (exact estimator), 32 m (network), 37 m (interpolation), against 17 m on the true map



lines every 400 m	**map rms**	**position rms**
true map	0	17.1 +- 0.7 m
tiled Wiener	0.63 nT	28.1 +- 4.9 m
network (38.5M)	0.71 nT	32.2 +- 5.6 m
linear interpolation	0.90 nT	37.0 +- 9.0 m

Survey lines every 400 m and 800 m, 75% of the map unobserved; each reconstruction is the navigation map against the true field. Three maps, 40 trajectories; the spread does not separate every pair.